

Exam 2

ECE 2201: Circuit Analysis I

Spring 2025

March 22, 2025

1. This exam is closed book, closed notes and a crib sheet has been provided to you with the exam. Print your name below.
2. Show all work on these pages. Show all work necessary to complete the problem and explain the steps in your solution. A solution without the appropriate work shown will receive no credit. A solution which is not given in a reasonable order will lose credit.
3. Show all units in solutions, intermediate results, and figures. Units in the exam will be included between square brackets.
4. If the grader has difficulty following your work because it is messy or disorganized, you will lose credit.
5. Do not use red ink. Do not use red pencil.
6. You will have 75 minutes to work on this exam.

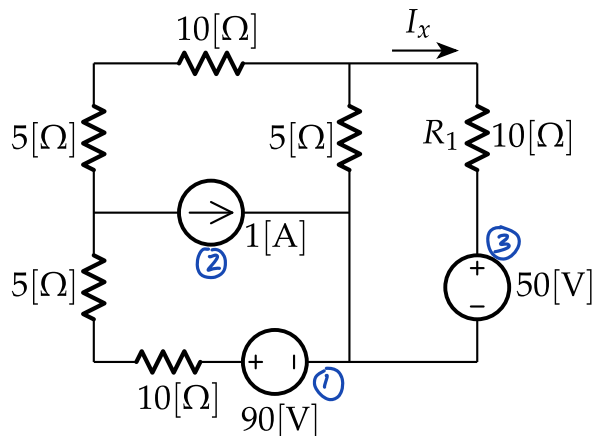
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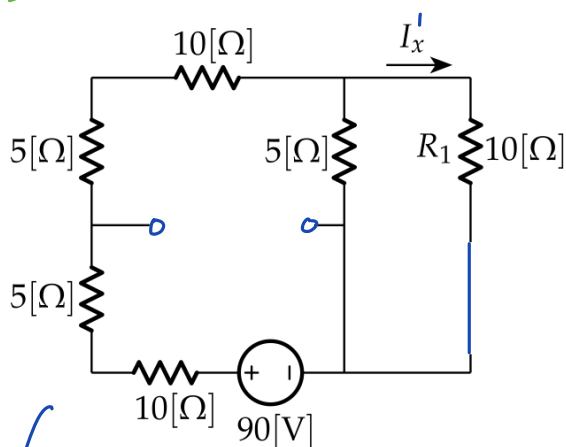
Problem 1

Note: to receive credit, you must use superposition to solve this problem.

- Use superposition to solve for I_x and write each solution for I_x' , I_x'' , I_x''' and I_x in the spaces below
- How much power is dissipated in R_1 ?



Source 1:



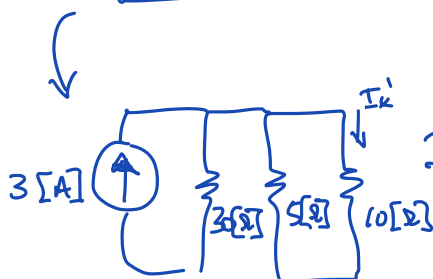
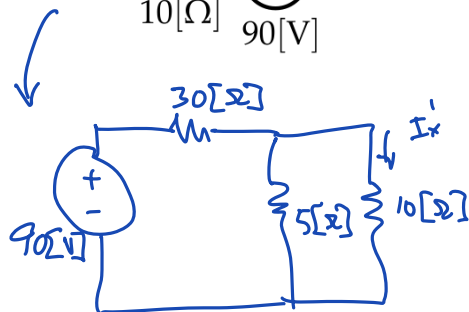
Alternative: node-voltage

$$\frac{V_x - 90[V]}{30[\Omega]} + \frac{V_x}{5[\Omega]} + \frac{V_x}{10[\Omega]} = 0$$

$$V_x - 90[V] + 6V_x + 3V_x = 0$$

$$10V_x = 90[V] \Rightarrow V_x = 9[V]$$

$$I_{x'} = \frac{V_x}{10[\Omega]} = \frac{9}{10}[A]$$



current divider

$$I_{x'} = \frac{\frac{30[\Omega]}{7}}{10[\Omega] + \frac{30}{7}[\Omega]} \cdot 3[A]$$

$$= \frac{30}{100} \cdot 3[A] = \frac{9}{10}[A]$$

$$30[\Omega] || 5[\Omega] = \frac{30 \cdot 5}{35} = \frac{30}{7}[\Omega]$$

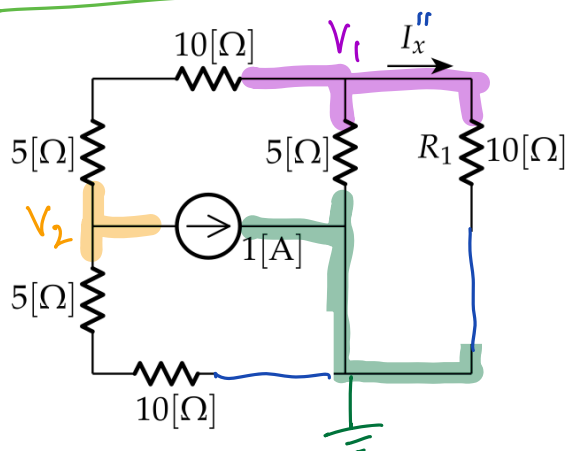
$$I_{x'} = \frac{9}{10}[A] = 0.9[A]$$

$$I_{x''} = -\frac{3}{20}[A] = -0.15[A]$$

$$I_{x'''} = -\frac{7}{2}[A] = -3.5[A]$$

$$I_x = -\frac{11}{4}[A] = -2.75[A]$$

$$p_{R1} = \frac{605}{8}[W] = 75.6[W]$$

Source 2

Node Voltage:

$$1[A] + \frac{V_2}{15[\Omega]} + \frac{V_2 - V_1}{15[\Omega]} = 0 \quad \times 15[\Omega]$$

$$15[V] + V_2 + V_2 - V_1 = 0$$

$$-V_1 + 2V_2 = -15[V] \quad (1)$$

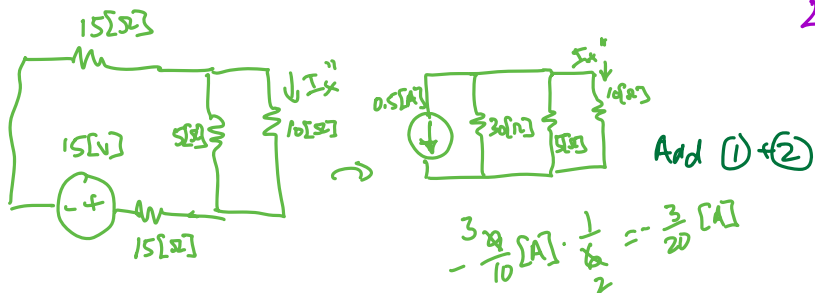
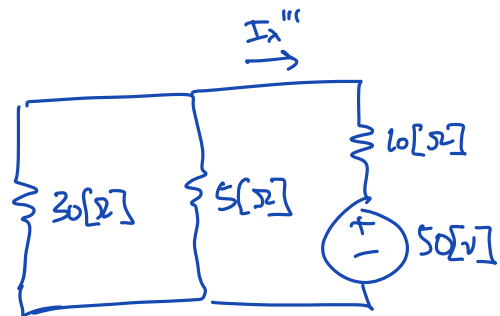
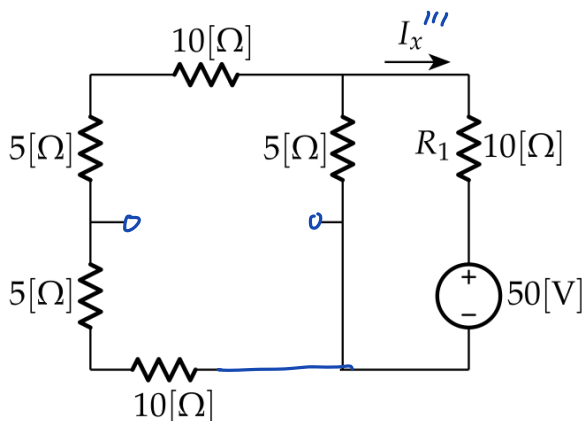
$$\frac{V_1 - V_2}{15[\Omega]} + \frac{V_1}{5[\Omega]} + \frac{V_1}{10[\Omega]} = 0 \quad \times 30[\Omega]$$

$$2V_1 - 2V_2 + 6V_1 + 3V_1 = 0$$

$$11V_1 - 2V_2 = 0 \quad (2)$$

$$10V_1 = -15[V] \Rightarrow V_1 = -1.5[V]$$

$$I_{x''} = \frac{V_1}{R_1} = \frac{-1.5[V]}{10[\Omega]} = -\frac{3}{20}[A]$$

Source 3:

$$I_{x'''} = \frac{-50[V]}{10[\Omega] + 30[\Omega] \parallel 5[\Omega]}$$

$$I_x = I_{x'} + I_{x''} + I_{x'''} \\ = -\frac{55}{20}[A] = -\frac{11}{4}[A] = -2.75[A]$$

$$P_R = I_x^2 \cdot 10[\Omega]$$

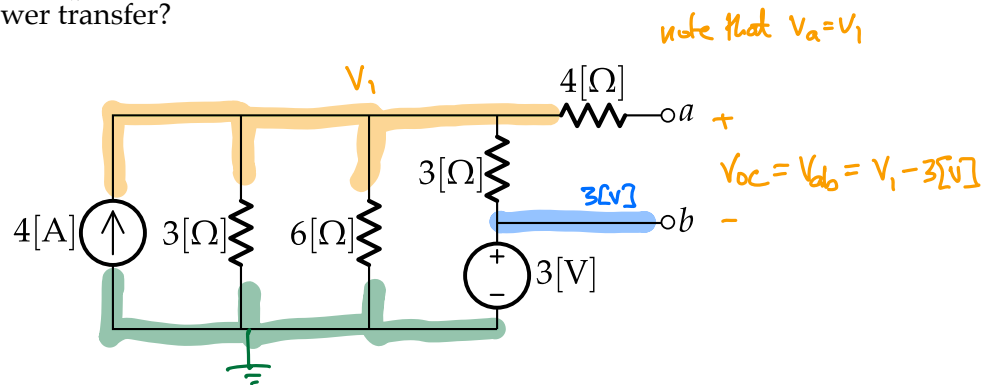
$$= \frac{121}{8} \cdot \frac{5}{16} [W] = \frac{605}{8} [W] = 75.6 [W]$$

$$30[\Omega] \parallel 5[\Omega] = \frac{30 \cdot 5}{35} [\Omega] \\ = \frac{30}{7} [\Omega]$$

$$I_{x'''} = \frac{-50[V]}{10[\Omega] + \frac{30}{7}[\Omega]} = \frac{-50}{2 \frac{100}{7}} [A] = -\frac{7}{2} [A]$$

Problem 2

- Find the Thévenin equivalent circuit at terminals (a,b) in the circuit below.
- If a load resistor is placed between terminals (a,b), what resistance value would ensure maximum power transfer?



open-circuit voltage

$$\left[4[A] = \frac{V_1}{3[\Omega]} + \frac{V_1}{6[\Omega]} + \frac{V_1 - 3[V]}{3[\Omega]} \right] \times 6[\Omega]$$

$$24[V] = 2V_1 + V_1 + 2V_1 - 6[V]$$

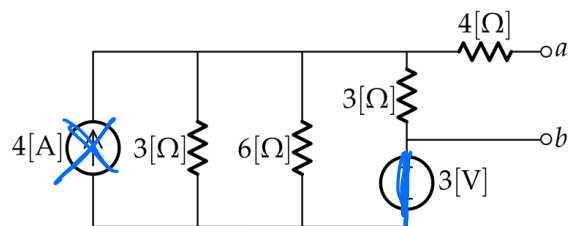
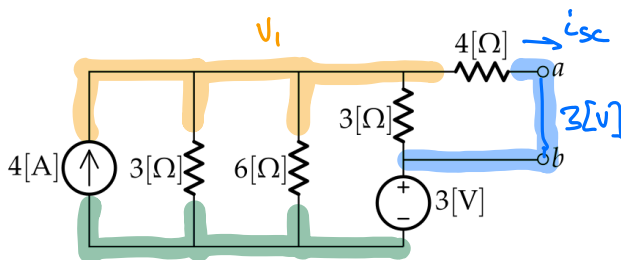
$$5V_1 = 30[V]$$

$$V_1 = 6[V]$$

$$V_{oc} = V_{Th} = V_1 - 3[V] = 3[V]$$

Find I_{sc} :

or use equiv. resistance



$$\left[4[A] = \frac{V_1}{3[\Omega]} + \frac{V_1}{6[\Omega]} + \frac{V_1 - 3[V]}{3[\Omega]} + \frac{V_1 - 3[V]}{4[\Omega]} \right] \times 24[\Omega]$$

$$96[V] = 8V_1 + 4V_1 + 8V_1 - 24[V] + 6V_1 - 8[V]$$

$$26V_1 = 138[V]$$

$$V_1 = \frac{138}{26} [V] = \frac{69}{13} [V]$$

$$I_{sc} = \frac{\frac{69}{13} [V] - 3[V]}{4[\Omega]} = \frac{\frac{30}{13}}{4} [A] = \frac{15}{26} [A]$$

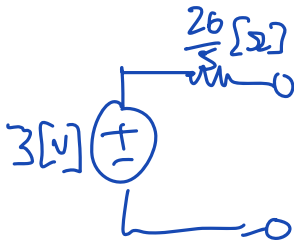
$$R_{Th} = \frac{V_{oc}}{I_{sc}} = \frac{3[V]}{\frac{15}{26} [A]} = \frac{26 \cdot 3}{15} [\Omega] = \frac{26}{5} [\Omega]$$

$$R_{eq} = 3[\Omega] \parallel 6[\Omega] \parallel 3[\Omega] + 4[\Omega]$$

$$\frac{\frac{3}{2} \cdot 6}{\frac{3}{2} + 6} [\Omega] = \frac{3 \cdot 6}{3 + 12} [\Omega] = \frac{3 \cdot 6}{15} [\Omega] = \frac{6}{5} [\Omega]$$

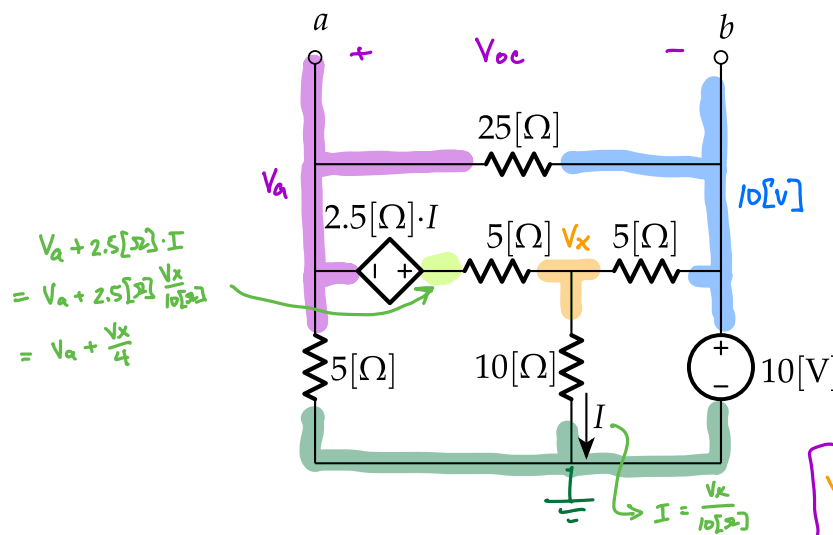
So:

$$R_{eq} = \frac{6}{5} [\Omega] + 4[\Omega] = \frac{6 + 20}{5} [\Omega] = \frac{26}{5} [\Omega] = 5.2[\Omega]$$



Problem 3

Find the Thévenin equivalent circuit at terminals (a,b) for the circuit below



$$\left[\frac{V_a - 10[V]}{25[\Omega]} + V_a + \frac{V_x}{4} - V_x + \frac{V_a}{5[\Omega]} = 0 \right] \times 100[\Omega]$$

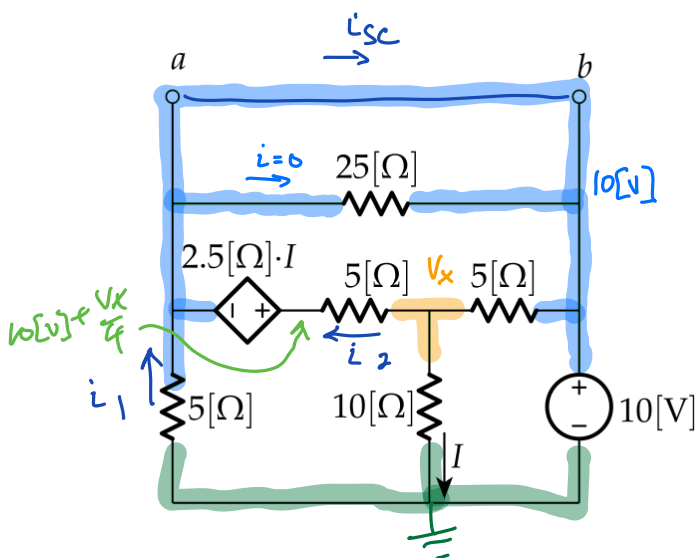
$$\begin{bmatrix} 44 & -15 \\ -4 & 9 \end{bmatrix} \begin{bmatrix} V_a \\ V_x \end{bmatrix} = \begin{bmatrix} 40 \\ 40 \end{bmatrix}$$

$$V_a = \frac{20}{7} [V] \quad V_x = \frac{40}{7} [V]$$

$$= 2.86 [V] \quad = 5.71 [V]$$

$$\begin{aligned} V_{bc} &= V_a - 10[V] = V_{th} \\ &= \frac{20}{7}[V] - \frac{70}{7}[V] = -\frac{50}{7}[V] = -7.14[V] \\ V_{th} &= -\frac{50}{7}[V] = -7.14[V] \end{aligned}$$

$$\left[\frac{V_x - (V_a + \frac{V_x}{4})}{5 \text{ [}\Omega\text{]}} + \frac{V_x - 10 \text{ [V]}}{5 \text{ [}\Omega\text{]}} + \frac{V_x}{10 \text{ [}\Omega\text{]}} = 0 \right] \quad + 20 \text{ [V]}$$



$$\frac{V_X - (10[V] + \frac{V_K}{4})}{5[S_2]} + \frac{V_X}{10[S_2]} + \frac{V_X - 10[V]}{5[S_2]} = 0$$

$$\left[\frac{\frac{3V_R}{4} - 10[V]}{5[\Omega]} + \frac{V_x}{10[\Omega]} + \frac{V_x - 10[V]}{5[\Omega]} = 0 \right] \times 20[\Omega]$$

$$3V_x - 40[V] + 2V_x + 4V_x - 40[V] = 0$$

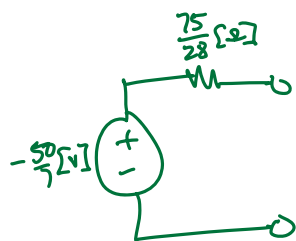
$$9 V_X = 80 [V]$$

$$V_X = \frac{80}{9} [V] = 8.89 [V]$$

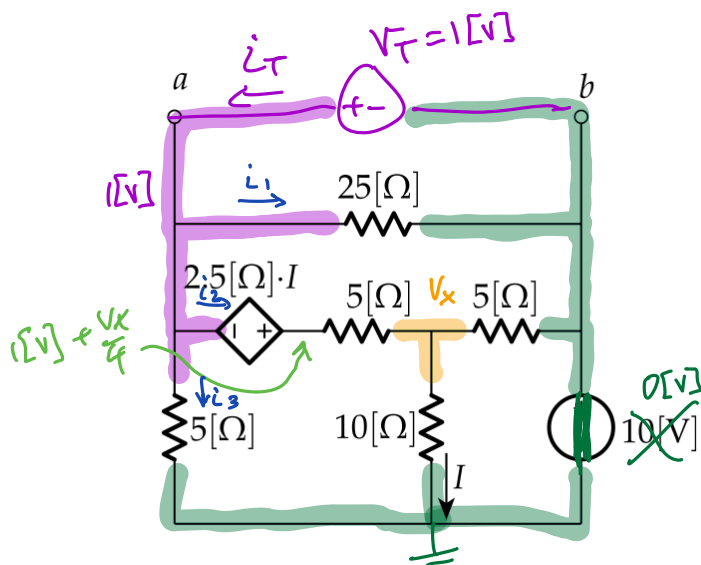
$$I_{sc} = I_1 + I_2 = \frac{-10[V]}{5[\Omega]} + \frac{V_X - (10[V] + \frac{V_X}{4})}{5[\Omega]}$$

$$= -2[A] + \frac{\frac{3V_x}{4} - 10[A]}{5[A]} = -\frac{8}{3}[A] = -2.67[A]$$

$$R_{Th} = \frac{V_{oc}}{I_{sc}} = \frac{-\frac{50}{7} [V]}{-\frac{8}{3} [A]} = \frac{25}{7} \cdot \frac{3}{8} [A] = \frac{75}{28} [A] = 2.68 [A]$$



Alternative to finding R_{Th} : use test source:



$$\frac{V_x - (1 [V] + \frac{V_x}{4})}{5 [A]} + \frac{V_x}{10 [A]} + \frac{V_x}{5 [A]} = 0$$

$$\left[\frac{\frac{3V_x}{4} - 1 [V]}{5 [A]} + \frac{V_x}{10 [A]} + \frac{V_x}{5 [A]} = 0 \right] \times 20 [A]$$

$$3V_x - 4 [V] + 2V_x + 4V_x = 0$$

$$9V_x = 4 [V]$$

$$V_x = \frac{4}{9} [V]$$

$$I_T = I_1 + I_2 + I_3 = -\frac{3}{4} V_x = -\frac{3}{4} \cdot \frac{4}{9} = -\frac{1}{3} [V]$$

$$= \frac{1 [V]}{25 [A]} + \frac{1 [V] + \frac{V_x}{4} - V_x}{5 [A]} + \frac{1 [V]}{5 [A]}$$

$$= \frac{1}{25 [A]} \left(1 [V] + 5 [V] - 5 \cdot \frac{3}{4} \cdot \frac{4}{9} [V] + 5 [V] \right)$$

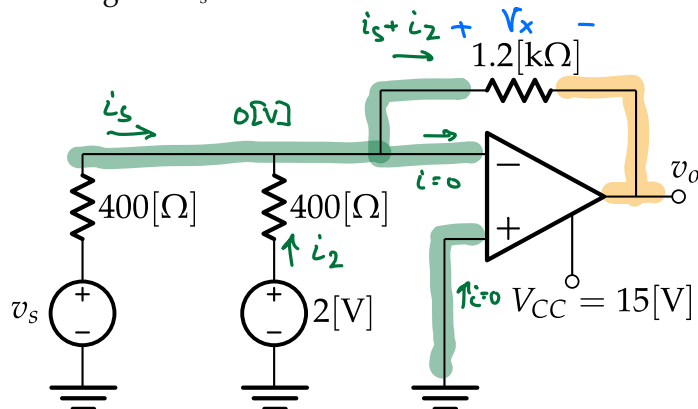
$$= \frac{1}{25 [A]} \left(11 [V] - \frac{5}{3} [V] \right)$$

$$= \frac{28}{75} [A]$$

$$R_{Th} = \frac{1 [V]}{\frac{28}{75} [A]} = \frac{75}{28} [A]$$

Problem 4

- Determine the output voltage (v_o) in the circuit below in terms of v_s .
- Specify the linear range for v_s .



Approach 1: inspection

$$\begin{aligned}
 V_x &= 1.2[\text{k}\Omega] (i_s + i_2) \\
 &= 1.2[\text{k}\Omega] \left(\frac{v_s}{400[\Omega]} + \frac{2[\text{V}]}{400[\Omega]} \right) \\
 &= 3v_s + 6[\text{V}]
 \end{aligned}$$

$$0[\text{V}] - v_o = V_x \Rightarrow v_o = -V_x$$

$$v_o = -3v_s - 6[\text{V}]$$

Approach 2: KCL

$$\begin{aligned}
 \frac{0 - v_s}{400[\Omega]} + \frac{0 - 2[\text{V}]}{400[\Omega]} + \frac{0 - v_o}{1200[\Omega]} &= 0 \\
 -3v_s - 6[\text{V}] - v_o &= 0
 \end{aligned}$$

$$v_o = -3v_s - 6[\text{V}]$$

$$-15[\text{V}] < v_o < 15[\text{V}]$$

$$-15[\text{V}] < -3v_s - 6[\text{V}] < 15[\text{V}]$$

$$-15[\text{V}] < 3v_s + 6[\text{V}] < 15[\text{V}]$$

$$-21[\text{V}] < 3v_s < 9[\text{V}]$$

$$-7[\text{V}] < v_s < 3[\text{V}]$$

